



BACKINT SDK for SAP HANA

1.50.2

Target Audience

- Development Partners

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1 Document History

The SDK described in this document is related to *BACKINT for SAP HANA* version 1.50

1.00

- Initial version

1.01

- changed internal formatting

1.02

- changed internal formatting

1.03

- changed internal formatting

1.04

- documentation for new test `InquireWrongSID` added

1.05

- documentation for new tests `InquireName2`, `InquireNameAndEBIDNeverMade` and `InquireName-AndEBIDDeleted` added

1.06

- documentation for new tests `ChangedDestinationWithEBID`, `ChangedDestinationWithNULL`, `ChangedDestinationWithEBIDNotFoundWrongSID`, `ChangedDestinationWithNULLNotFoundWrongSID` and `ChangedDestinationNotFoundWrongEBID` added

1.07

- migrated document source from Doxygen to LaTeX

1.50

- Documentation for backint 1.5
- New test cases : `BackupTypes`, `MissingDBNameInSIDParameter`
- New cmd parameter : `BACKINT_VALIDATOR_BACKINT_PROTOCOL_VERSION`

1.50.1

- Update formatting

1.50.2

- Adapt installation instructions to shipment in HANA server package

2 Introduction

2.1 Overview

This document describes the *BACKINT SDK for SAP HANA*. The goal of this toolkit is to enable 3rd party backup tool vendors to implement their *BACKINT for SAP HANA*¹ and thus allow the use of their backup solution in combination with the SAP HANA database.

BACKINT for SAP HANA (shortened as *HDBBACKINT* for the rest of this document) is an executable which must be implemented by a 3rd party backup tool vendor. When a database backup is started this executable is in turn called by the database. *hdbbackint* has to execute all activities needed to transfer the backup data to a secure storage. The data to be backed up is read by the backup tool from pipes which are created in advance by the database.

Besides this documentation, the *BACKINT SDK for SAP HANA* contains documentation on the API of *BACKINT for SAP HANA*, an executable named *hdbbackintvalidator.sh* to test your *hdbbackint* implementation and a suite of basic tests

¹For the specification of the API please refer to the documentation "Backint Specification" which is part of this Toolkit.

3 hdbbackintvalidator

hdbbackintvalidator is an interactive tool which can be used to test your *hdbbackint* implementation.

3.1 Prerequisites

hdbbackintvalidator is only supported on SLES and RHEL systems with Linux x86-64 and PowerPC architectures where HANA is delivered. And since it simulates backups and recoveries it creates files of different sizes on your file system. Thus, it is essential, that there is enough disk space available which depends on with which backup sizes you will test with. (The backup sizes can be set by a command line option. Additionally you can cleanup your disk using the tool in between the tests.)

To run the tool a user other than root should be used.

3.2 Installation of the hdbbackintvalidator

To install the *hdbbackintvalidator* first need to install the SAP HANA server software using the `hdb1cm` command.

You will see output like this when running the installation:

```
...
Preparing package 'Backint Validation Tools'...
...
Installing package 'Backint Validation Tools'...
...
```

To use the tool, the `hdbbackintvalidator_pack.tar.gz` file must be unpacked using the `tar` command. The `hdbbackintvalidator_pack.tar.gz` file is located in the directory `/usr/sap/<SID>/exe/hdb/sdk/backint`. It is advised that you unpack the archive into a directory of your choice.

```
tar -xvzf /usr/sap/<SID>/exe/hdb/sdk/hdbbackintvalidator_pack.tar.gz
```

These directories and files will be unpacked to a location of your choice.

```
...
hdbbackintvalidatortestingcertification.sh
hdbbackintvalidator.sh
lib/
lib/hdbbackintvalidator
...
```

Additionally your implemented *hdbbackint* executable must be located in a directory contained in the `PATH` environment variable.

3.3 Starting the hdbbackintvalidator

After you unpacked it and placed your hdbbackint executable under \$PATH, you can run the tool without any command line options as follows:

```
yourHost:/someWhere/hdbbackintvalidator $ ./hdbbackintvalidator.sh
```

You will get an output containing a list of tests for you to execute.

You select the function you would like to test by typing the corresponding number and hitting enter. For example, if you would like to execute a simple data backup (SimpleBackup) you select “1” and a more detailed selection offers the decision between different backup variants (e. g. SimpleBackup. Then you select “1” again to execute the simple data backup.

```
...

==== initial setUp:
      initialSetUp called for 50 test cases
Select the test to execute:
  1: Backup (SimpleBackup, TwoBackupsInSamePipe, ParallelBackup,
      ParallelDataAndLogBackup, VerySmallBackup, VeryLargeBackup,
      BackupFileNames, BackupUsingFileSpec, BackupTypes)
  2: Delete (SimpleDelete, DeleteTwo, DeleteTwoReverse, NotFound1,
      NotFound2, NotFoundPartiallyCorrect, OneOfTwoNotFound)
  3: E2E (TopoBackup, DataBackup, CatBackup, CatRestore, TopoRestore,
      DataRestore)
  4: Error (ErrorPipeNotFound, ErrorPipeNotPermitted, ErrorPipeClosed,
      ErrorInputFileNotFound, ErrorInputFileNotPermitted,
      ErrorInputFileNotValid, ErrorOutputFileNotFound,
      ErrorOutputFileNotPermitted, ErrorEBIDNotFound,
      ChangedDestinationWithEBIDNotFoundWrongSID,
      ChangedDestinationWithNULLNotFoundWrongSID,
      ChangedDestinationNotFoundWrongEBID, MissingDBNameInSIDParameter)
  5: Inquire (InquireName, InquireName2, InquireNameAndEBID,
      InquireNameAndEBIDNeverMade, InquireNameAndEBIDDeleted,
      InquireLatest, InquireWrongSID, InquireNameDeleted)
  6: Other (PrintBackintVersion)
  7: Restore (SimpleRestore, GenerationRestore, ParallelRestore,
      ChangedDestinationWithEBID, ChangedDestinationWithNULL)
  8: ZZCleanup (ZZCleanup)
  0: all tests
  t: set/print tracelevel
  c: set/print checklevel
  e: endless test run      (current = false)
  m: malfunction run      (current = false)
  p: print parameters
  o: option menu
  x: exit

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Backup
- 1: SimpleBackup
```



```

- 2: TwoBackupsInSamePipe
- 3: ParallelBackup
- 4: ParallelDataAndLogbackup
- 5: VerySmallBackup
- 6: VeryLargeBackup
- 7: BackupFileNames
- 8: BackupUsingFileSpec
- 9: BackupTypes
  0: all subtests
  x: exit

```

1

If the test finished successfully, this will be reported accordingly on the console.

```

==== Backup: Starting, Seed = 1844884233
==== Trace is written to: /tmp/ngdb_test_XYZ/exe/test_run
==== Other files are written to: /tmp/ngdb_test_XYZ/exe/test_run
==== TestCase SetUp
---- SetUp
backintRestorePath: /tmp/ngdb_test_XYZ/exe/test_run/ChangedDest/
backintBackupPath: /tmp/ngdb_test_XYZ/exe/test_run/
backintParameterFile:
backintToolOption:
---- Test case start: SimpleBackup
Create simple backup with size: 104857600
Create file:
    BackintValidator_1397466726534_104857600_SimpleBackupRestore
Start backint
Start copy to backint
Progress: 16777216/18446744073709551615
Progress: 33554432/18446744073709551615
Progress: 50331648/18446744073709551615
Progress: 67108864/18446744073709551615
Progress: 83886080/18446744073709551615
Progress: 100663296/18446744073709551615
Progress: 104857600/18446744073709551615
Finished copy
Finished backint
Backint Version: backint 1.50 SoftwareVersion: HANA backup agent
    2.50.00.01.000000
name: /tmp/ngdb_test_XYZ/exe/test_run/SimpleBackupRestore, rc: OK, EBID:
    BCKINTw0R5ai
IBID = [1397466726534] EBID = [BCKINTw0R5ai] pipe = [/tmp/ngdb_test_XYZ/
    exe/test_run/SimpleBackupRestore] backupSize = 104857600
---- CleanUp/TearDown
---- run time SimpleBackup: 2843323 usec (brutto), 2839895 usec (netto)
---- Test case successful: SimpleBackup
==== TestCase TearDown
---- Executed 1/1 tests successfully
==== Backup OK
==== TOTAL: successfully executed 1/1 tests
==== Test(s) finished successfully

```

The output contains additional information about the progress of the test and the steps it went through. In our example you are told, that initially ...

```
==== Trace is written to: /tmp/ngdb_test_XYZ/exe/test_run
==== Other files are written to: /tmp/ngdb_test_XYZ/exe/test_run
```

The test creates files named `BackintValidator_<IBID>_<backupsizesize>_<pipename>` of the size `<backupsizesize>` in this directory simulating a SAP HANA database backup file, like

```
Create file: BackintValidator_47110070815_104857600_SimpleBackupRestore
```

Then the test starts your `hdbbackint` and writes this file to a pipe named `SimpleBackupRestore` and from the pipe to another file named somehow. In this example it is called `<pipename>.<EBID>`:

```
SimpleBackupRestore.P.XYZ.BCKINTw0R5ai
```

3.4 Available Tests

If you use the `hdbbackintvalidator` with the command line option `-l` (see below for the most relevant command line options) you get a list of available tests.

```
yourHost:/someWhere/hdbbackintvalidator # ./hdbbackintvalidator.sh -l
Backup: SimpleBackup
Backup: TwoBackupsInSamePipe
Backup: ParallelBackup
Backup: ParallelDataAndLogbackup
Backup: VerySmallBackup
Backup: VeryLargeBackup
Backup: BackupFileNames
Backup: BackupUsingFileSpec
Backup: BackupTypes
Delete: SimpleDelete
Delete: DeleteTwo
Delete: DeleteTwoReverse
Delete: NotFound1
Delete: NotFound2
Delete: NotFoundPartiallyCorrect
Delete: OneOfTwoNotFound
E2E: TopoBackup
E2E: DataBackup
E2E: CatBackup
E2E: CatRestore
E2E: TopoRestore
E2E: DataRestore
Error: ErrorPipeNotFound
Error: ErrorPipeNotPermitted
Error: ErrorPipeClosed
Error: ErrorInputFileNotFound
Error: ErrorInputFileNotPermitted
Error: ErrorInputFileNotValid
Error: ErrorOutputFileNotFound
Error: ErrorOutputFileNotPermitted
Error: ErrorEBIDNotFound
Error: ChangedDestinationWithEBIDNotFoundWrongSID
Error: ChangedDestinationWithNULLNotFoundWrongSID
Error: ChangedDestinationNotFoundWrongEBID
Error: MissingDBNameInSIDParameter
Inquire: InquireName
Inquire: InquireName2
Inquire: InquireNameAndEBID
Inquire: InquireNameAndEBIDNeverMade
Inquire: InquireNameAndEBIDDeleted
Inquire: InquireLatest
Inquire: InquireWrongSID
Inquire: InquireNameDeleted
Other: PrintBackintVersion
Restore: SimpleRestore
Restore: GenerationRestore
Restore: ParallelRestore
```

```
Restore:ChangedDestinationWithEBID
Restore:ChangedDestinationWithNULL
ZZCleanup:ZZCleanup
```

This section will give an overview of the existing tests.

3.4.1 Backup

Backup tests first create one or more files simulating the SAP HANA database backup files. This file's size can be set by passing the command line option "-P BACKUP_SIZE" to the *hdbbackintvalidator*. Then the test starts *hdbbackint*, writes this file to a pipe and afterwards writes it out of the pipe to your *hdbbackint* backup file.

Subtest	Description
Simple Backup	Creates a dummy backup file, starts <i>hdbbackint</i> and writes this file to a pipe. <i>hdbbackint</i> should receive the backup data from the pipe and store it in a new <i>hdbbackint</i> backup file.
TwoBackupsInSamePipe	Creates two dummy backup files, starts <i>hdbbackint</i> and writes these files into the same pipe. <i>hdbbackint</i> should receive the backup data from the pipe and store it in two new <i>hdbbackint</i> backup files.
ParallelBackup	Creates two dummy backup files, starts <i>hdbbackint</i> and writes these files in parallel into two pipes. <i>hdbbackint</i> should receive the backup data from these pipes and store them in parallel in two <i>hdbbackint</i> backup files.
ParallelDataAndLogbackup	Creates dummy data and log backup files, starts <i>hdbbackint</i> and writes them to pipes. <i>hdbbackint</i> should receive the backup data from these pipes and store them in <i>hdbbackint</i> data and log backup files.
VerySmallBackup	Creates a small backup (4 KB = 4096 B)
VeryLargeBackup	Creates a large backup (1.1GB = 1073741824 B). If the command line option "-P BACKUP_SCALE" is used (see section below) the default values of 1 GB for this test can be further scaled by the given factor.
BackupFileNames	Use backup names containing special characters.
BackupUsingFileSpec	Backup and Recopvery using a file specification instead of a pipe specification
BackupTypes	Creates backups with different backup types that are passed to <i>hdb-backint</i> with -l parameter

3.4.2 Delete

In order to support housekeeping and the cleanup of incomplete or erroneous backups the function *delete* has to be supported.

Subtest	Description
SimpleDelete	Creates a backup and deletes it
DeleteTwo	Creates two backups and deletes them with a single backing call
DeleteTwoReverse	Creates two backups and deletes them with two backing calls, the first one deleting the second backup created
NotFound1	Attempts to delete a backup that does not exist
NotFound2	Attempts to delete two backups that do not exist with a single backing call
NotFoundPartiallyCorrect	Creates a backup named "2bdeleted". Attempts to delete a backup with that name and an EBID that does not exist. Attempts to delete a backup with a name that does not exist and the EBID of the backup "2bdeleted". Finally deletes the backup "2bdeleted"
OneOfTwoFound	Creates two backups. Deletes the second one.. Then attempts to delete both backups with a single backing call.

3.4.3 E2E

This test simulates a sequence of backups and recoveries of a HANA database instance performing a data backup and a data recovery: The data backup saves the topology, the data of several services in parallel and the backup catalog. Each save task is performed by a distinct backing call, but all save tasks share the same session id. The data recovery restores the backup catalog, the topology and the data itself.

Subtest	Description
TopoBackup	Simulates the metadata backup
DataBackup	Simulates the parallel data backups of two services
CatalogBackup	Simulates the backup catalog backup
CatalogRestore	Simulates the backup catalog restore
TopoRestore	Simulates the topology restore
DataRestore	Simulates the parallel data restore"

3.4.4 Error

These tests check that for certain error situations a valid error message is generated.

Subtest	Description
ErrorPipeNotFound	Execute simple restore from pipe -> Start <i>hdb-backint</i> : ERROR: System error: /tmp/ngdb_test/exe/test_run/TestError, rc=2: No such file or directory
ErrorPipeNotPermitted	Executes simple restore from pipe -> start <i>hdb-backint</i> : ERROR: System error: /tmp/ngdb_test/exe/test_run/TestError, rc=13: Permission denied
ErrorPipeClosed	Executes a backup to a pipe that is closed before all data has been transmitted
ErrorInputFileNotFound	start <i>hdbbackint</i> : ERROR: Cannot open input file '/var/tmp/hdbbackint.49IkS7'
ErrorInputFileNotPermitted	start <i>hdbbackint</i> : ERROR: Cannot open input file '/var/tmp/hdbbackint.sBnBys'
ErrorInputFileNotValid	start <i>hdbbackint</i> : ERROR: Invalid line '#PIPE' found in <i>hdbbackint</i> communication file
ErrorOutputFileNotFound	Start <i>hdbbackint</i> : ERROR: Cannot open output file '/Schneewittchen/BeiDen7Zwergen/un-sinn.txt'
ErrorOutputFileNotPermitted	Start <i>hdbbackint</i> : ERROR: Cannot open output file '/var/tmp/hdbbackint.Kj3Ozx'
ErrorEBIDNotFound	Start <i>hdbbackint</i> : Error reported by <i>hdbbackint</i> : 'BACKINT recovery job into /tmp/ngdb_test/exe/test_run/TestError failed with error code 2' in file '/var/tmp/hdbbackint.0DvB5k'
ChangedDestinationWithEBIDNotFoundWrongSID	Start <i>hdbbackint</i> for a recovery with changed destination, correct EBID but wrong system id. Error reported by <i>hdbbackint</i> : Backint cannot find file '/tmp/ngdb_test_XYZ/exe/test_run/ChangedDestinationWithEbidNotFound-WrongSID'
ChangedDestinationWithNULLNotFoundWrongSID	Start <i>hdbbackint</i> for a recovery with changed destination, #NULL as EBID and a wrong system id. Error reported by <i>hdbbackint</i> : Backint cannot find file '/tmp/ngdb_test_XYZ/exe/test_run/ChangedDestinationWithNullNotFound-WrongSID'
ChangedDestinationNotFoundWrongEBID	Start <i>hdbbackint</i> for a recovery with changed destination and wrong EBID. Error reported by <i>hdbbackint</i> : Backint cannot find file '/tmp/ngdb_test_XYZ/exe/test_run/Changed-DestinationNotFoundWrongEBID'
MissingDBNameInSIDParameter	Start <i>hdbbackint</i> with only SID passed to -u parameter instead of DBNAME@SID. <i>hdbbackint</i> should return an error

3.4.5 Inquire

The function *inquire* is used to obtain information from the backup server. An input file contains information about the items for which information is requested and returns the result to an output file.

Subtest	Description
InquireName	inquire with name and #NULL specified
InquireName2	inquire with two entries with name and #NULL specified
InquireNameAndEBID	inquire with name and EBID specified
InquireNameAndEBIDNeverMade	inquire with name and EBID specified. The specified backup has never been made. Backint should return NOTFOUND.
InquireNameAndEBIDDeleted	inquire with name and EBID specified. The specified backup has been made and deleted. Backint should return NOTFOUND.
InquireLatest	inquire without anything specified. The SID used is taken from \$SAPSYSTEMNAME. Backint should return a list of backups made starting with the latest one.
InquireWrongSID	inquire without anything specified. The SID used is different from \$SAPSYSTEMNAME. Backint should return an empty list.
InquireNameDeleted	create two backups into the same pipe, delete one of them and inquire the remaining with only the pipe name specified.

3.4.6 Other

Additionally to the API tests some other tests are provided.

Subtest	Description
PrintBackintVersion	Prints the version of <i>hdbbackint</i>

3.4.7 Restore

These tests first create one or more files simulating the SAP HANA database backup files. These files are read into a pipe and *hdbbackint* creates a *hdbbackint* backup file. This file is restored again using a pipe.

Subtest	Description
SimpleRestore	Creates one dummy backup file, starts <i>hdbbackint</i> and writes it to a pipe. <i>hdbbackint</i> should receive the backup data from the pipe, store them in <i>hdbbackint</i> backup files and restore them using a pipe.
GenerationRestore	Creates two backups with the same pipe name , then restores the first one (with EBID specified) and after that the second one (without EBID specified)
ParallelRestore	Creates two dummy backup file, starts <i>hdbbackint</i> and writes them to two pipes. <i>hdbbackint</i> should receive the backup data from the pipes, store them in <i>hdbbackint</i> backup files and restore them using a pipe.
ChangedDestinationWithEBID	Creates a backup and restores it using a different pipe as used for backup. The name of the pipe used for recovery is the same as for backup but it is located in the directory specified by the parameter RESTORE_PATH. The backup is fully specified by the pipe, ebid and the system id
ChangedDestinationWithNULL	Creates a backup and restores it using a different pipe as used for backup. The name of the pipe used for recovery is the same as for backup but it is located in the directory specified by the parameter RESTORE_PATH. The backup is specified by the pipe, #NULL as wildcard for the ebid and the system id

3.4.8 ZZCleanup

A pseudo-test that deletes all files created by previous tests

Subtest	Description
ZZCleanup	Deletes all files created by previous tests

3.5 Test protocol

After the test a test protocol named *hdbbackint*validator_<yyyymmdd>_<hhmmss>.txt (<yyyymmdd> and <hhmmss> replaced by date and time of the start of *hdbbackint*validator) is created in the current directory. It contains information about the date and time and the version of the *hdbbackint* used and a summary of the test results.

4 hdbbackintvalidator - Commandline Options

When using the hdbbackintvalidator you also can set some options and start tests from the command line. Here are some of the most commonly used are described in more detail.

Option	Description	Comment
-h	Offers some helpful information	
-l	Lists the available test cases	
-A	Run all available test cases	
-P BACKUP_PATH=<directory>	Allows the user to specify a directory for the backups	Enough disk space for the backups must be available. If this parameter is not specified, backups are written to the directory /tmp/ngdb_test/exe/test_run
-P RESTORE_PATH=<directory>	Allows the user to specify a directory in which the pipes for recovery are created in testcases which uses different directories for backup and restore.	If this parameter is not specified, pipes for recovery are created in the directory /tmp/ngdb_test/exe/test_run/ChangedDest
-P BACKUP_SIZE=<size>	Creates backups of this size	Enough disk space for the backups must be available
-P BACKUP_SCALE=<factor>	This option BACKUP_SCALE expects a value (a factor) by which the 1 GB backup size of the VeryLargeBackup test are scaled.	Only to be used with VeryLargeBackup test.
-P PARAMETERFILE=<file>	This option is used to forward a special parameter file into each hdbbackint execution request.	
-P TOOLOPTION=<toolOptions>	This option is used to provide some special tool options which are forwarded using the #TOOLOPTION in the input file.	
-P BACKINT_VALIDATOR_BACKINT_PROTOCOL_VERSION=<version>	This option is used to set the backint protocol version used by the hdbbackintvalidator.	Possible options are 1.0 and 1.5. Default value is 1.5
testname[:subtest]	Run single tests	You can select tests from the list received from option "-l", e. g. Backup:Simple-Backup
testname	Run all tests belonging to testname	For example selecting "Backup" will start all subtests of this testname one after the other.